

Attn-CapBN: An Attention-based Bangla Image Captioning using the Bangla View Dataset

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Abstract

Understanding images through natural language descriptions is a fundamental challenge in artificial intelligence, bridging computer vision and natural language processing. While significant progress has been made in English image captioning, resource-scarce languages such as Bangla remain underexplored. To address this gap, this study introduces BanglaView, a large-scale Bangla image captioning dataset inspired by Flickr30k, consisting of 31,783 images paired with 158,915 professionally verified Bangla captions. Each image is annotated with five diverse and fluent descriptions, ensuring both quality and linguistic richness. Building on this resource, we propose Attn-CapBN, an encoder–decoder framework that employs a custom-designed CNN feature extractor and a visual attention mechanism with a GRU decoder to generate contextually coherent Bangla captions. Rigorous training and evaluation were conducted using both BanglaView and the BAN-Cap dataset. Experimental results demonstrate that Attn-CapBN achieves strong performance, with BLEU-1 to BLEU-4 scores of 0.623, 0.487, 0.394, and 0.333 on BanglaView, and 0.620, 0.485, 0.398, and 0.332 on BAN-Cap, respectively. These results surpass existing baselines, highlighting the effectiveness of the proposed CNN-based feature extractor and attention-guided decoding. The contributions of this work include the release of BanglaView as a benchmark dataset and the introduction of Attn-CapBN as a robust architecture for Bangla image captioning, paving the way for further advancements in regional language captioning and multimodal AI research.

Keywords: Bangla Image Captioning, Machine Translation, Attn-CapBN, Convolutional Neural Networks Model, Data Processing and Analysis, Bangla View Dataset.

1. Introduction

Image captioning bridges computer vision and natural language by using CNNs to extract visual features and NLP models to generate descriptions. For example, Asraful et al. [1] developed a lightweight convolutional neural network to recognize handwritten Bengali characters and numerals, achieving state-of-the-art accuracy on multiple Bangla character datasets. In general

captioning frameworks, a CNN encoder is paired with an RNN or Transformer decoder to produce word sequences that describe an image. Large annotated datasets and novel modeling techniques have driven much progress. Hodosh et al. [2] introduced the Flickr8k dataset (8,000 images with five human captions each) and proposed framing image description as a ranking task. They used kernel canonical correlation analysis (KCCA) to build strong baseline systems for image-caption retrieval and showed that ranking-based evaluation metrics align better with human judgments than traditional BLEU or ROUGE. Young et al. [3] later developed denotational similarity metrics by constructing a hierarchy (denotation graph) of visual concepts from 30K images and 150K captions, improving semantic inference over event descriptions. Meanwhile, Lin et al. [4] presented the MS COCO dataset with 328K images (2.5M object instances) and rich object segmentation annotations, pushing captioning models to handle complex everyday scenes. Researchers also extended captioning to multiple languages. Funaki and Nakayama [5] proposed an image-mediated learning approach for zero-shot cross-lingual document retrieval. They use images as a “hub” to link languages, learning a shared semantic space for English and Japanese via a generalized CCA on triplets of (image, English text, Japanese text), and released a trilingual dataset of image-text-document pairs. Elliott et al. [6] released Multi30K, which augments Flickr30k images with German translations of English captions (by professionals) and independently crowdsourced German descriptions. Grubinger et al. [7] introduced the IAPR TC-12 benchmark of 20,000 images with multilingual captions (English, German, Spanish), enabling evaluation of visual information systems in multiple languages. These efforts show how combining visual and linguistic data across languages facilitates tasks like multilingual captioning and cross-lingual retrieval.

Despite Bengali being one of the world’s most spoken languages (over 230 million speakers), captioning resources for Bangla have been scarce [8]. Mansoor et al. [9] created *BanglaLekhaImageCaptions*, a culturally relevant dataset of 9,154 Bangladeshi images each annotated with two Bengali captions by native speakers. Building on this, Rahman et al. [10] developed Chittron, a Bangla captioning model that integrates a pre-trained VGG16 CNN (for image encoding) with stacked LSTM layers (for caption decoding). They collected 16,000 Bangladeshi images with Bangla annotations and trained the VGG16+LSTM network to generate captions word-by-word. This model demonstrated the feasibility of Bangla captioning, achieving good BLEU scores on their dataset. Deb et al. [11] proposed Oboyob, a Bengali captioning engine that adapts standard CNN-LSTM architectures. They subsampled and preprocessed a machine-translated Bengali version of the Flickr8K captions, then trained various CNN-LSTM models (using merge and inject fusion architectures) on these data. Oboyob evaluated 16 model variants and established baseline performance for Bengali captioning, revealing the limitations of current word embeddings for Bangla. More recently, Shah et al. [12] introduced Bornon, a transformer-based captioning model for Bangla. Unlike RNN decoders, their Transformer processes captions without strict sequence order. They trained it on three Bengali image-caption datasets and showed that the transformer approach outperforms a visual-attention encoder-decoder baseline. They also

collected 4,000 images with five Bengali captions each (albeit relatively short ones), and demonstrated superior metrics for the Transformer model compared to prior CNN-RNN models.

In summary, each prior work provides complementary technical advances: CNN architectures for Bangla character and scene understanding [1], ranking and evaluation methods for image-sentence alignment [2,3], large-scale annotated datasets [4–7], and specialized Bangla captioning models (CNN+LSTM [10,11], Transformer [12]). Yet existing Bangla datasets are limited in size or caption richness. To address this, we introduce BanglaView: a new Bengali image-caption dataset built from Flickr30k’s 31,783 images, annotated with 158,915 Bengali sentences (five per image) by bilingual native speakers. The annotations were carefully reviewed and post-processed to ensure linguistic and cultural quality. This resource, along with our captioning methods, closes the gap in Bangla multimodal research and enables further technical advancements in multilingual image captioning. So, we have created a model Attn-CapBN for generating descriptions of images in the Bengali language by utilizing the encoder-decoder framework with an attention mechanism. The encoder component is based on convolutional neural networks (CNN) and is responsible for extracting relevant features from the images. On the other hand, the decoder component is based on gated recurrent units (GRU) and is responsible for generating accurate descriptions for the images. The main contributions of this are as follows:

- The attention-based encoder-decoder model Attn-CapBN is presented in this study for the purpose of captioning Bangla images.
- To learn deep visual representations from images, we develop a novel CNN feature extractor that does not take pretrained models into account.
- In order to dynamically focus on significant portions of the image when writing a caption, this research developed a visual attention tool.
- An attention-weighted decoder based on GRU is used in this work to generate Bangla captions that are fluent and delicate to context.
- This study introduces BanglaView, a new dataset for human-verified annotations used to caption images in Bangla.
- We evaluate model performance using standard metrics like BLEU, METEOR, ROUGE-L, and CIDEr on two datasets: BanglaView and BAN-Cap.

2. Literature Review

This section delves into the advancements made in image captioning, highlighting the diverse research endeavours and model development efforts that have culminated in the generation of syntactically accurate captions.

Hossain et al. [13] proposed a Bangla image captioning approach using EfficientNetB4 and ResNet-50 for feature extraction, introducing the Bangla View dataset alongside Flickr30k. Their experiments showed that EfficientNetB4 achieved superior results with a BLEU score of 0.54 after only 10 epochs, demonstrating its effectiveness in generating coherent Bangla captions and advancing research in accessibility and digital communication. Palash et al. [14] proposed a

CNN+Transformer architecture with visual attention. Specifically, they used a pre-trained ResNet-101 image encoder and a Transformer-style decoder. On the BanglaLekhaImageCaptions dataset, this model set a new benchmark for Bangla captioning: BLEU-1 = 0.694, BLEU-2 = 0.630, BLEU-3 = 0.582 (and METEOR 0.337), far surpassing earlier Bengali models. This confirms that combining CNN visual features with Transformer attention yields strong results on Bangla data. Baidya et al. [15] presented “Bangla Chitra Net,” a sequence-to-sequence model using a Transformer encoder-decoder. Their architecture takes raw images (sequentialized) as input, encodes them via a CNN-based image encoder, and decodes with an RNN+attention to output Bengali captions. They evaluated on both the BAN-Cap dataset and Bengali-translated Flickr8k, and report that their Transformer-based model outperforms previous CNN+Transformer approaches. Masud et al. [16] introduced a dual-CNN attention model for Bengali captions. Their end-to-end system uses *two separate CNNs* to extract image features, fusing them with attention before decoding into Bangla text. Trained on a Bengali version of Flickr8k (the BAN-Cap dataset), this model achieved BLEU-4 = 0.430, which the authors cite as “the pinnacle of current attainable standards”. Thus, employing multiple CNN feature extractors with attention substantially improves Bengali captioning.

Bhuiyan et al. [17] proposed a context-aware attention mechanism for Bengali image captioning using a ResNet-50 encoder and a bidirectional GRU decoder. Their model effectively captured contextual dependencies, outperforming traditional semantic attention approaches. Experiments on BAN-Cap, BanglaLekhaImageCaption, and Bornon datasets showed significant METEOR score improvements, highlighting the potential of attention-based methods in low-resource languages. Das et al. [18] developed an encoder–decoder model for Bengali image captioning, addressing the lack of standard datasets in this language. They curated errors in two available Bengali datasets and benchmarked their model, which outperformed baseline approaches. The system achieved BLEU-1 scores of 0.67 and 0.65, and BLEU-4 scores of 0.26 and 0.24, highlighting progress in regional vision–language understanding. Das et al. [19] proposed an encoder–decoder model for Bengali image captioning, using a dataset with two descriptions per image. Their work addressed the scarcity of standard Bengali datasets and demonstrated the model’s effectiveness, achieving a BLEU-1 score of 0.67. They emphasized that larger and more diverse datasets could further enhance captioning performance in low-resource languages. Rahman et al. [20] also used a ResNet-50 encoder with a bidirectional LSTM decoder. Trained on BanglaLekhaImageCaptions, CapNet achieved very high BLEU scores (BLEU-1 = 0.81, BLEU-4 = 0.51). The authors noted that experimenting with different pre-trained feature extractors (VGG16, Xception, etc.) validated ResNet-50 as the most effective for Bangla captioning. This high BLEU-1 suggests that larger Bangla image datasets allow CNN+LSTM models to produce very fluent captions. Naim et al. [21] presented “Bornomala,” an attention-based CNN–LSTM captioning model. They extract CNN object features and apply attention in the decoder LSTM. Evaluated qualitatively, they reported BLEU scores between 0.35 and 0.44, indicating reasonable caption quality. This again demonstrates that a hybrid CNN+LSTM+attention pipeline can learn to caption images in Bengali.

This research introduces Attn-CapBN, an encoder–decoder framework with a visual attention mechanism for Bangla image captioning. The encoder employs a custom CNN to extract

meaningful image features, while the decoder uses GRUs to generate fluent and contextually accurate Bangla descriptions. A new dataset, BanglaView, along with the existing BAN-Cap dataset, is utilized for training and evaluation. Comparative analysis between machine-generated and human-verified captions, supported by statistical insights, is presented. Experimental results show that Attn-CapBN outperforms state-of-the-art models across BLEU, METEOR, ROUGE-L, and CIDEr metrics, advancing Bangla image captioning research.

3. The BanglaView Dataset

3.1 Data Collection

We based our image source on the Flickr30k captioning benchmark [3]. Flickr30k consists of 31,783 images collected from the Flickr photo-sharing platform, covering a wide range of everyday scenes and activities. Each image is annotated with five English captions that were obtained via crowdsourcing. The BanglaView dataset then provides Bangla translations for these captions. In total, BanglaView contains 31,783 images and 158,915 English–Bangla caption pairs, i.e. five Bangla captions per image. The Bangla captions were generated by native Bengali speakers fluent in English, and a professional review team carefully post-processed all captions to ensure linguistic accuracy and consistency.

3.2 Data Processing

We employed a multi-step annotation pipeline to translate and verify the captions. The main steps of our data processing were:

- **Automated Translation.** All 158,915 English captions were first translated into Bangla using Google Translate. The raw translations were then manually reviewed to correct obvious errors and improve fluency.
- **Tiered Human Verification.** To ensure quality, annotators were organized into multiple tiers. A first group of native Bangla linguists performed the initial review of the translated captions. A second group of bilinguals (Bangla–English) graduate students then examined these annotations for accuracy and consistency. During this process, each Bangla sentence was compared against its source image and English caption to verify that the description remained faithful to the visual content.
- **Final Proofreading.** Finally, a third group of Bangla language experts performed a last proofreading pass to catch any remaining grammatical or spelling mistakes. This multi-stage review process—translation by machine followed by successive expert refinements—mirrors protocols used in prior Bangla captioning datasets.

Each step of this pipeline was designed to minimize translation artifacts and ensure that the final Bangla captions accurately describe the images in fluent, idiomatic Bangla.

3.3 Machine Translation vs Human-Verified Data

Even though automated machine translation programs such as Google Translator have advanced significantly, they still perform poorly in many areas, particularly when it comes to low-resource

languages like Bangla. The following significant elements become evident from a side-by-side comparison as important arguments in favour of human-verified dataset curation as opposed to machine-translated content:

At the moment, automated translators are not sufficiently tailored for Bangla. The structural integrity of a Bangla sentence is preserved in just a small percentage of machine-translated captions, notwithstanding their incoherence with the picture image content. Fig. 1 illustrates this discovery visually.

	English Caption A man and woman looking at a baby in a room with a clock on the wall. Google Translated Caption দযোলে ঘড়ি লাগিয়ে একটি ঘরে একটি শিশুর দকি তাকিয়ে একজন পুরুষ এবং মহিলা Human Verified Caption দযোলে ঘড়ি লাগানো একটি রুমে একটি শিশুর দকি তাকিয়ে আছে একজন পুরুষ এবং মহিলা
	English Caption Man on a bicycle riding on only one wheel. Google Translated Caption একটি সাইকেলে চড়ে মানুষ শুধুমাত্র একটি চাকার উপর চড়ে Human Verified Caption একজন মানুষ একটি সাইকেলে শুধুমাত্র একটি চাকার উপর চালাচ্ছে
	English Caption There are many people in a large room with a lot of furniture with tags on them. Google Translated Caption একটা বড় কক্ষে অনেকে লোক আছে যাদের গায়ে ট্যাগ লাগানো অনেকে আসবাবপত্র Human Verified Caption একটা বড় কক্ষে অনেকে লোক ট্যাগ লাগানো অনেকে আসবাবপত্র দেখছে

Fig. 1. Several examples of machine-translated captions that can adequately convey the image's context, but fall short of preserving the syntactical and structural integrity of a Bangla phrase.

Machine-translated Bangla captions sometimes lack meaningful coherence due to a high frequency of misspelt words, improper punctuation, and missing phrases. Fig. 2 provides a clear illustration of this problem.

	English Caption Three people reading books. Google Translated Caption বই পড়ছে তনিজন Human Verified Caption তনিজন ব্যক্তি বই পড়ছে
	English Caption People sitting inside a bus waiting to reach their destination. Google Translated Caption গন্তব্যে পৌঁছানোর অপেক্ষায় বাসরে ভতেরে বসে থাকা লোকজন Human Verified Caption গন্তব্যে পৌঁছানোর লক্ষ্যে বাসরে ভতেরে কিছু লোক বসে অপেক্ষা করছে
	English Caption Person in hat painting a multi level wooden fence. Google Translated Caption একটি বিহু স্তরের কাঠের বড়ো পইন্টিং টুপি ব্যক্তি Human Verified Caption টুপি পিরা একজন ব্যক্তি একটি বিহু স্তরের কাঠের বড়ো রঙ করছে

Fig. 2. Several examples of machine-translated captions that fall short of accurately describing the image or, in certain situations, fail to produce any coherent statement at all.

It's common for machine-translated subtitles to miss cultural subtlety. It is clear from the examples in Fig. 1 and Fig. 2 that the system translates the source language almost entirely word for word into the target language. The Bangla captions also contain a significant amount of transliterated English terms, which results in a loss of syntactic and semantic sense. When compared to their machine-generated counterparts, the human-verified captions demonstrate a wide range of variation. In our case, the data that was machine-translated had 25,950 distinct tokens, but the data that was checked by humans had 26,328 tokens more than this. For the reasons described above, captions produced by systems trained on machine-translated data frequently have an unnatural quality and are not applicable in real life.

3.4 Statistical Analysis of the BanglaView Dataset

Comprehensive corpus-level statistics and a comparison between BanglaView and other available datasets in the Bangla language are shown in Table 1. BanglaView has a greater number of unique tokens than other datasets, making it perform better. Although BanglaLekhaImageCaptions [9] has a similar average phrase length, BanglaView has more than twice as many captions. Interestingly, the recently released Bornon [12] human-annotated dataset shows a significantly shorter average phrase length, which is important for collecting fine details when describing pictures. One noteworthy finding is that BanglaView's English and Bangla captions differ structurally from one another. To be more precise, Bangla descriptions in BanglaView have almost 87% more unique tokens than English descriptions. In contrast, the entire quantity of Bangla tokens.

The top 20 most frequent tokens in the Flickr30k English caption and BanglaView Bangla caption datasets are shown graphically in Fig. 3 and Fig. 4 respectively. It is interesting to observe that start and stop words occur rather often in both languages. Our research removes these start and stop words to guarantee a relevant comparison. Interestingly, the token frequencies in both datasets continue to be very similar even when these components are removed. Here are some common stop words in Bangla: এ, অথবা, হয়, করা, হয়েছে, যে, এই, এবং, হল, সে, তার, তাদের, তারা , এক ,একটি, একটা, যা, যার, যাদের, কোন.

Table 1. Statistics of the BanglaView dataset's textual data combined with the data currently in existence for the captioning of Bangla images

Dataset Name	#Sentences	#Total Tokens	#Unique Tokens	Sentence Length Mean	Sentence Length Variance
Flickr8k (English) [3]	40455	437421	8440	10.81	14.51
BAN-Cap (Bangla) [33]	40455	344574	15846	8.51	10.99
BLIC (Bangla) [10]	18308	155249	5720	8.47	20.13
Bornon (Bangla) [34]	20500	110566	6228	5.34	4.38
Chittron (Bangla) [11]	16000	160000	15000	10	
BNLIT (Bangla) [22]	8743	87430	15000	10	
Flickr30k (English) [4]	158915	1943528	20124	12.23	27.02
BanglaView (Bangla)	158915	1668322	26348	10.49	20.82

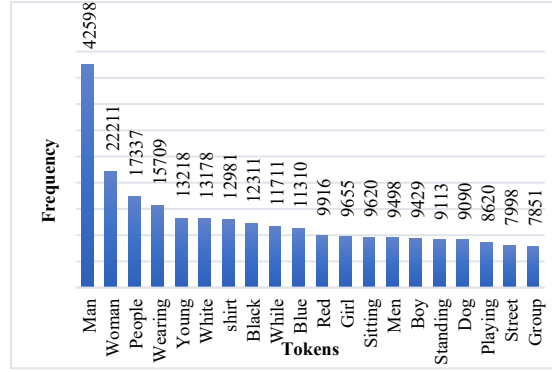


Fig. 3. Flickr30k dataset most frequently used English tokens



Fig. 4. BanglaView dataset most frequent Bangla tokens (a) excluding stop words (Left) (b) including stop words (Right)

4. Methodology

In this section, we present the methodology underlying our proposed image captioning model. At the core of our framework lies a novel Convolutional Neural Network (CNN) designed for robust image feature extraction. Building upon this, the architecture adopts an encoder–decoder structure enhanced with a visual attention mechanism. Specifically, the encoder integrates a CNN with an attention module to selectively focus on salient regions of the input image, thereby capturing contextually significant visual cues. The decoder, implemented using gated recurrent units (GRU), sequentially generates Bangla captions by leveraging both the extracted features and the attention-driven context vectors. This attention-guided encoder–decoder paradigm, illustrated in Fig. 5, enables the Attn-CapBN model to dynamically align visual features with linguistic elements, resulting in more accurate and context-aware descriptions. The detailed hyperparameter configurations employed during training are also provided to ensure reproducibility of the proposed approach.

4.1 System Architecture

The proposed model, Attn-CapBN, follows an encoder–decoder framework integrated with a visual attention mechanism to generate Bangla image captions. As illustrated in Fig. 5, the architecture is divided into two main components: the encoder and the decoder. In the encoder, image features are first extracted using a CNN model and passed through a fully connected layer followed by ReLU activation to obtain a dense feature representation. These features are then processed by the attention mechanism, which dynamically emphasizes salient regions of the image, thereby producing a context vector that captures the most informative visual cues. The context vector is subsequently combined with the hidden state and word embeddings to form a focused visual–semantic representation. In the decoder, a GRU network utilizes this enriched context to sequentially generate Bangla word tokens [22]. These tokens are further passed through fully connected layers and a softmax classifier to produce the final caption. This attention-guided encoder–decoder paradigm enables Attn-CapBN to align visual features with linguistic elements more effectively, resulting in fluent and context-aware Bangla captions.

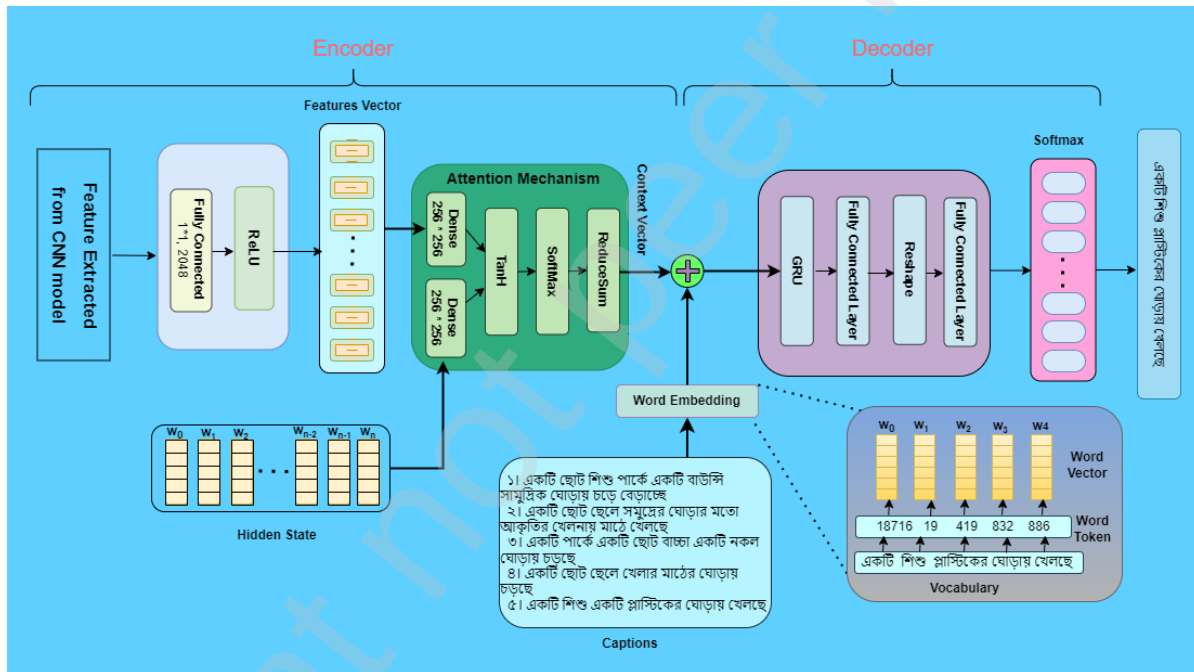


Fig. 5. System architecture of the proposed Attn-CapBN model, an attention-based encoder–decoder framework for Bangla image captioning.

4.1.1 Attention Mechanism

The core novelty of Attn-CapBN lies in its attention module, which adaptively directs focus toward different parts of the image when generating each word of the caption. As shown in Fig. 6, the model highlights specific visual regions (e.g., the baby and the plastic horse) to guide caption generation. The mechanism begins by computing an alignment score between the encoder outputs and the decoder’s previous hidden state:

$$Alignment_{value} = w_{alignment} \cdot \tanh(w_{decoder} \cdot H_{decoder} + w_{encoder} \cdot H_{encoder}) \dots \dots \dots (1)$$

Where, $W_{alignment}$ is the trainable vector, $W_{decoder}$ is the weighted value of the decoder, $H_{decoder}$ is the hidden state value of the decoder, $W_{encoder}$ is the weighted value of the encoder, $H_{encoder}$ is the hidden state value of the encoder.

As the encoder output and decoder hidden state go through the complex network architecture, they pass through several Linear layers that are each outfitted with a different set of trainable weights. After the alignment scores vector is created in the previous phase, it is necessary to apply a softmax function to this vector to convert it and produce the attention weights. The equation for this:

$$a_{ij} = Softmax(Alignment_{value}) \dots \dots \dots (2)$$

After the attention weights have been computed in the previous step, we may create the context vector by multiplying the attention weights by the encoder outputs elementally.

$$con_t = \sum_{j=1}^{Tx} (a_{ij} \cdot h_j) \dots \dots \dots (3)$$

Where con_t is the context vector.

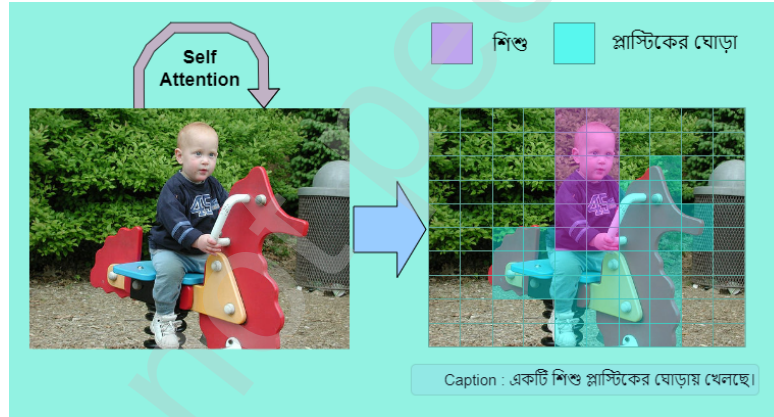


Fig. 6. Attention on an Image.

4.1.2 CNN Feature Extraction Network

Fig. 7 illustrates the design of the CNN feature extractor developed for our proposed Attn-CapBN model. Unlike many prior works that rely on pretrained backbones, our CNN extractor is designed from scratch to capture rich visual representations tailored for Bangla image captioning. The architecture consists of five sequential stages:

- **Stage 1:** Two consecutive bottleneck blocks with 128 filters each, followed by a 1×1 convolutional layer with stride 2 and a 3×3 max-pooling layer with stride 2.

- **Stage 2:** Two consecutive bottleneck blocks with 128 filters, and a third block with 256 filters. Each block applies pre-activation units, where batch normalization and activation precede convolution.
- **Stage 3:** Three consecutive bottleneck blocks, with 256 filters in the first two blocks and 512 filters in the third block.
- **Stage 4:** Four consecutive bottleneck blocks, with 512 filters in the first two blocks and 1024 filters in the third block.
- **Stage 5:** Five consecutive bottleneck blocks, with 1024 filters in the first two blocks and 2048 filters in the third block.

The custom-built CNN architecture comprises 30.6 million parameters, enabling it to effectively capture complex patterns and features from the input data. This feature extractor ensures effective hierarchical feature learning, enabling the encoder to generate high-quality visual embeddings for the attention mechanism in Attn-CapBN.

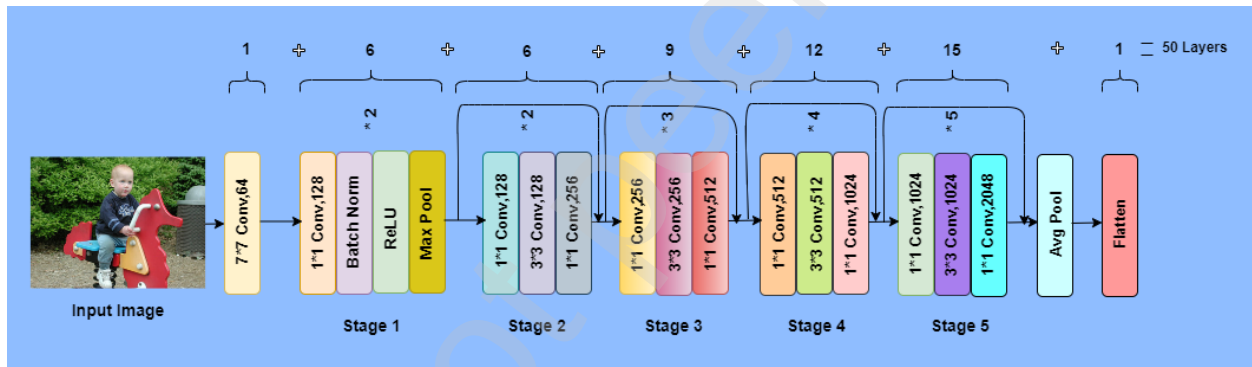


Fig. 7. Architecture of the custom CNN feature extractor used in the Attn-CapBN model.

4.2 Hyperparameters

Table 2 shows the various hyper-parameter values that we used to train Attn-CapBN model. During the training phase, our model modifies those internal variables to reduce the error between the predicted and target captions. The properties, relationships, and patterns of the data are all fully captured by those model parameters.

Table 2. Experimental parameters for BanglaView dataset.

Parameters name	Value	Parameters name	Value
Vocabulary size	5000	Test images	6357
Batch size	64	Dimension of model	512
Buffer size	1000	Number of Epoch	50
Train images	25426	Maximum length of sentence	65

5. Result Analysis

This section presents a comparative analysis of Attn-CapBN model's performance with other widely recognized models in the context of image captioning. For detailed results, please refer to Table 3.

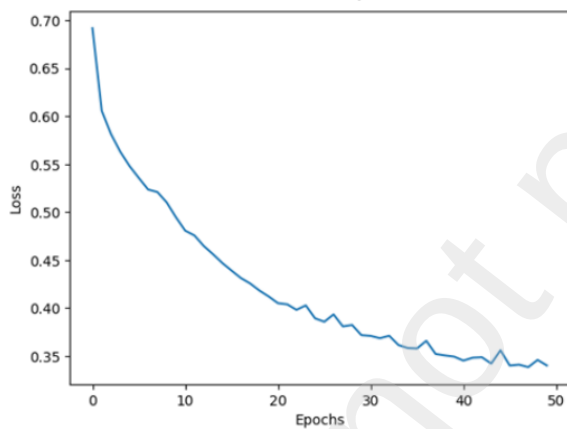
Table 3. Showcasing the performance of each model on the BanglaView and BAN-Cap datasets.

Model name	Dataset	BLEU 1	BLEU 2	BLEU 3	BLEU 4	METEOR	ROUGE	CIDEr
CNN-Merge based [23]	BAN-Cap	0.565	0.355	0.221	0.131	0.281	0.290	0.178
Visual-Attention based [24]	BAN-Cap	0.587	0.368	0.254	0.144	0.293	0.288	0.199
Transformer [12]	BAN-Cap	0.623	0.396	0.251	0.152	0.300	0.290	0.198
InceptionV3 [25] + Proposed Attn-CapBN model	BAN-Cap	0.583	0.444	0.389	0.214	0.291	0.312	0.261
InceptionV3 [25] + Proposed Attn-CapBN model	BanglaView	0.550	0.418	0.337	0.286	0.294	0.274	0.222
EfficientNetV2L [26] + Proposed Attn-CapBN model	BanglaView	0.589	0.454	0.366	0.308	0.297	0.296	0.272
ConvNeXtXLarge [27] + Proposed Attn-CapBN model	BanglaView	0.612	0.480	0.386	0.326	0.309	0.322	0.323
ResNet50V2[28] + Proposed Attn-CapBN model	BAN-Cap	0.591	0.452	0.394	0.222	0.309	0.322	0.274
ResNet50V2[28] + Proposed Attn-CapBN model	BanglaView	0.609	0.481	0.384	0.322	0.308	0.322	0.324
Attn-CapBN Model	BAN-Cap	0.620	0.485	0.398	0.332	0.311	0.328	0.283
Attn-CapBN Model	BanglaView	0.623	0.487	0.394	0.333	0.311	0.323	0.324

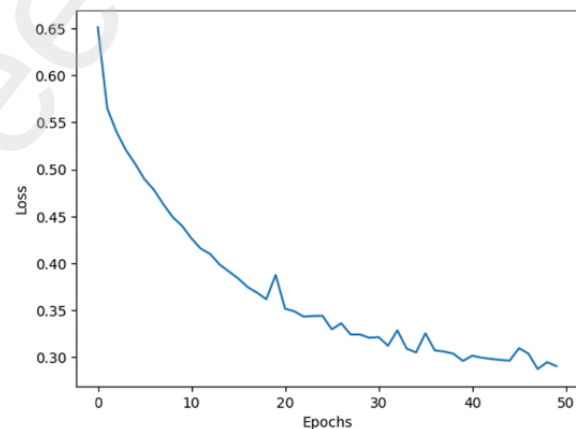
The models' performances were assessed using established evaluation metrics, including BLEU [28], METEOR [29], ROUGE-L[30], CIDEr[31], and. The geometric means of the n-grams in the reference and predicted phrases are evaluated to calculate BLEU. However, the ratings might not always be exact since there is variation in how the same language is represented with equal meaning. 1 to 4-gram BLEU values were used to evaluate the anticipated captions. The longest common subsequences between reference and forecasted phrases are examined to determine ROUGE-L. This measure finds the longest co-occurrence in n-grams between sentences while taking sentence-level structural similarity into account. A Term Frequency Inverse Document Frequency (TF-IDF) weighting method is used to the n-grams in each phrase to compute CIDEr. This method makes it easier to assess how relevant and related phrases are in the context of each

sentence. METEOR is calculated by first calculating the accuracy and recall harmonic means, then comparing the real and predicted sentences word for word. Metrics such as CIDER and SPICE [32] are specific to assessing the syntactic and semantic quality of picture captions, whereas BLEU and METEOR are used to evaluate machine translation and image captioning. On the test set of the main dataset, Table 4 shows the assessment scores of several picture captioning models. The CNN-Merge [23] model had the lowest scores among all assessment measures. Through the selective extraction of salient characteristics from a picture during caption prediction, the Visual-Attention model [24] shows a performance improvement. The Transformer model [12] also shows a good performance.

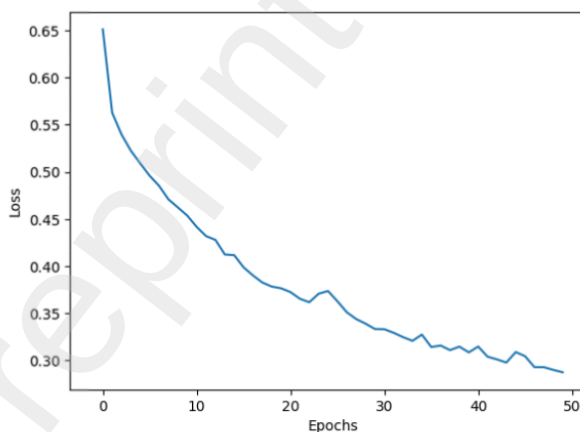
Using a visual sentinel to better direct the attention mechanism, the proposed model outperforms all other models in most assessment measures. Notably, adding text augmentation results in improved performance for all models. After a great deal of testing with different combinations of the text augmentation approaches, it is found that the proposed model, with Contextualized Word Replacement in particular, produces the best assessment ratings.



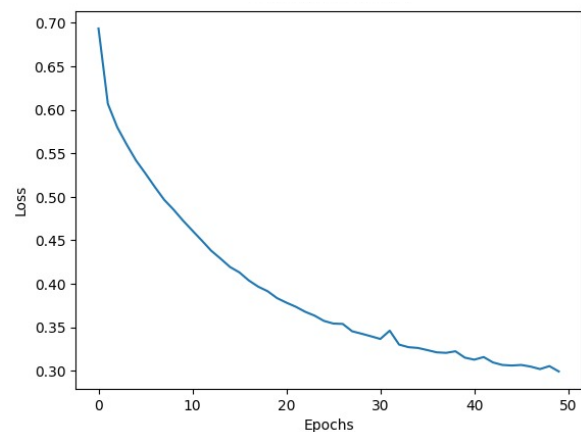
(a) Loss Plot: InceptionV3



(b) Loss Plot: EfficientNetV2L



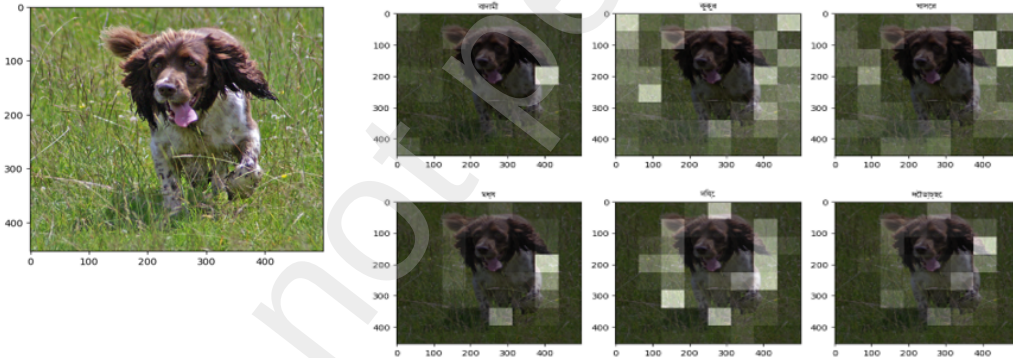
(c) Loss Plot: ResNet50V2



(d) Loss Plot: Proposed Attn-CapBN Model

Fig. 8. Loss curve of different models.

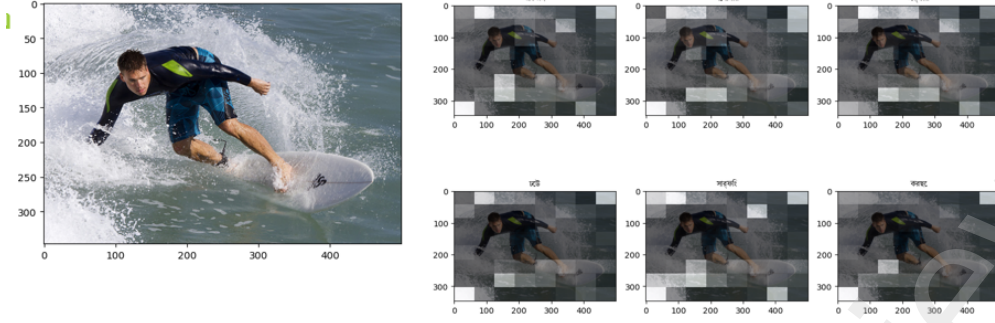
From Table 3 we can see that, using the BanglaView dataset the proposed Attn-CapBN model achieved BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGEL, and CIDEr scores of 0.623, 0.487, 0.394, 0.333, 0.311, 0.323, and 0.324 respectively. Using the BAN-Cap dataset the proposed Attn-CapBN model achieved BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGEL, and CIDEr scores of 0.620, 0.485, 0.398, 0.332, 0.311, 0.328, and 0.324 respectively. We further evaluated our model by employing several other cutting-edge CNN models, including InceptionV3, EfficientNetV2L, ConvNeXtXLarge, and ResNet50V2, and observed promising results. From the aforementioned discourse, it can be inferred that our proposed visual attention based Attn-CapBN architecture, which is based on an encoder-decoder framework and incorporates the custom convolutional neural network (CNN), exhibits superior performance when compared to the existing models such as CNN-Merge [23], Visual attention [24], and Transformer [12]. Here below Fig. 8 shows the loss plot of different models. The loss curve of our proposed Attn-CapBN model is smoothly decreasing according to increasing the number of epochs that's indicates a good model. Here Fig. 9 the predicted caption by our model is based on the focusing point of the image that is indicated by a white square box. A more lighted square box means highly important and less lighted means less important.



Predicted Caption: বাদামী কুকুর ঘাসের মধ্য দিয়ে দৌড়াচ্ছে

Real caption was:

- ১। খুব ফ্লপি কানসহ একটি বাদামী কুকুর ঘন ঘাসের মাঠের মধ্য দিয়ে চলে
- ২। কুকুরটি তার জিভ বের করে ঘাসে হাটছে
- ৩। বাদামী এবং সাদা কুকুর ঘাসের উপর দিয়ে ক্যামেরার দিকে দৌড়াচ্ছে
- ৪। একটি স্প্যানিয়েল ঘাসের মধ্য দিয়ে চলে
- ৫। ঘাসের মধ্যে একটি তুলতুলে কুকুর



Output Caption: একজন পেশাদার সারফার ডেউ সারফিং করছে

Real caption was:

- ১। একটি দীর্ঘ-হাতা শাট এবং বোর্ড শটস পরা একজন ব্যক্তি ক্যামেরার দিকে তাকিয়ে সারফ করছে এবং জলে তার ডান হাত দিয়ে ঝুঁকছে
- ২। একজন সারফার পানির গভীরে ডেউ কাটতে কাটতে সারফিং করছে
- ৩। সমুদ্র উপকূলে ছোট তরঙ্গের মধ্যে একজন সারফার
- ৪। সুটে পরা একজন ব্যক্তি নীল জলের মধ্যে ছোট তরঙ্গে সারফ করছে
- ৫। একজন লোক একটি সারফবোর্ডে সারফিং করছে

Fig. 9. Prediction image caption with attention plot by proposed model.

6. Conclusion

This work introduced Attn-CapBN, an attention-based encoder–decoder captioning model that uses a custom CNN feature extractor and a GRU decoder, and BanglaView, a professionally annotated English–Bangla image–caption dataset (31,783 images, 158,915 caption pairs) designed as a high-quality resource for multimodal and multilingual research. Language experts verified the dataset's annotations, which were then thoroughly examined statistically and qualitatively to show that they were appropriate for both machine translation and picture captioning tasks. In order to boost caption variety, text-augmentation techniques were investigated; preliminary testing suggests that these augmentations can strengthen the resilience of the model.

According to empirical evaluation on two Bangla benchmarks, BAN-Cap and BanglaView, the suggested model performs competitively. The model obtained BLEU-1 = 0.620 and BLEU-4 = 0.332 on BAN-Cap, and BLEU-1 = 0.623 and BLEU-4 = 0.333 on BanglaView. The efficacy of Attn-CapBN's CNN+attention architecture is demonstrated by comparative evaluation using BLEU, METEOR, ROUGE-L, and CIDEr. The model generates more accurate, fluid, and contextually matched Bangla captions than a number of previous baselines.

Future research will keep the bespoke CNN extractor as a dedicated visual backbone while examining transformer-based decoders and wider multimodal pretraining regimes. BanglaView is anticipated to support future developments in multimodal translation, regional language captioning, and downstream activities including image-based question formulation and assistive technology for visually impaired users due to its size, expert curation, and proven usefulness.

8. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

9. Data availability

The BanglaView dataset introduced in this research is now publicly available in the link below <https://data.mendeley.com/datasets/rrv8pbxr/v1>

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